

Context-Aware Speech Stress Detection in Hospital Workers Using Bi-LSTM Classifiers

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Abstract—Hospital workers are known to work long hours in a highly stressful environment. The COVID-19 pandemic has increased this burden multi-fold. Pre-COVID statistics already showed that one in every three nurses reported burnout, thus affecting patient satisfaction and the quality of their provided service. Real-time monitoring of burnout, and other underlying factors, such as stress, could provide feedback not only to the clinical staff, but also to hospital administrators, thus allowing for supportive measures to be taken early. In this paper, we present a context-aware speech-based system for stress detection. We consider data from 144 hospital workers who were monitored during their daily shifts over a 10-week period; subjective stress readings were collected daily. Wearable devices measured speech features and physiological readings, such as heart rate. Environment sensors, in turn, were used to track staff movement within the hospital. Here, we show the importance of context-awareness for stress level detection based on a bidirectional LSTM deep neural network. In particular, we show the importance of hospital location and circadian rhythm based contextual cues for stress prediction. Overall, we show improvements as high as 14% in F1 scores once context is incorporated, relative to using the speech features alone.

I. INTRODUCTION

Living in a changing environment with unprecedented challenges takes its toll on people's emotional and mental health. These challenges have increased recently due to the prevalence of COVID-19 across the world. This pandemic has changed people's lives and affected their lifestyle, both at home and at work. Most notably, it has put an additional burden on health care systems around the world, directly affecting patient outcomes and increasing nurse burnout rates [1]. With the prevalence of the COVID-19 pandemic, it is vital to ensure that health workers provide rapid, accurate, and satisfactory care services to patients [2].

Prior to COVID-19, statistics already showed that one in every three nurses reported suffering from burnout; these numbers increased multi-fold since the pandemic started [3]. Stressed hospital staff can result in lower patient satisfaction, poor staff well-being, and, ultimately, on reduced patient safety [4]. As such, it is crucial that systems be put in place to help detect burnout early, as well as many of its facets, including stress, mental and physical fatigue, mental workload, and compassion fatigue [5]. Wearable technologies have emerged as a promising tool for stress detection [6].

It is known that stress manifests itself in numerous ways in human psychophysiology. For example, alterations in heart

rate and heart rate variability have been reported [7], as have changes in breathing patterns [8], and muscle tension of the vocal cords, especially when performing a secondary physical task [9], [10]. Avila *et al.* [9], for example, showed that stressful states can lead to increased energies at higher acoustic frequencies, as well as a shift in the fundamental frequency f_0 , when compared to neutral speech. Others have also shown changes in speech jitter and in mel-frequency cepstral coefficients (MFCC) [11]. Portable audio solutions have been developed to allow for unobtrusive audio activity monitoring, thus opening the doors for wearable audio solutions [12].

Notwithstanding, context can play a crucial role on stress. Several studies have shown the effects that the surrounding environment can have on a person's stress level. For example, participants walked through different acoustic scenes and significant physiological changes were observed based on location and noise levels [13]. Moreover, people exposed to biophilic indoor environments exhibited lower blood pressure and lower stress levels [14]. Noise levels have been shown to affect stress levels [15], particularly in hospital workers [16].

Another contextual cue that has shown to be important for stress monitoring is that of circadian rhythm. This rhythm refers to 24-hour cycle that is observed in many physiological variables (e.g., heart rate [17]) driven by exogenous cycles of light and darkness, temperature, and sleep-wake cycle, to name a few [18]. Circadian rhythms have been reported to be an important factor in mental health, alertness and human performance [19], with disruptions causing significant effects on mood, processing speed, and working memory [20], as well as anxiety and depression [21]. Recent research has shown that this rhythm can be quantified with wearable devices [22], thus further pushing for a complete wearable solution.

In this paper, we use wearable multimodal data (physiological, location, and speech) collected from clinical staff of a large hospital in California during a 10-week period [23] to show the importance of context-awareness for stress level detection. In particular, we show the performance gains obtained when using location- and circadian rhythm awareness, coupled with a bidirectional long short-term memory (LSTM) neural network [24], relative to using speech features alone.

The rest of this paper is organized as follows: Section II describes the methods and material used. Section III presents the experimental results and a discussion. Finally, Section IV presents the conclusions and future work.

TABLE I
STUDY DEMOGRAPHICS.

Demographics	
Total Participants	212
Study Duration	10 weeks
Age	38.5 ± 9.94
Female / Male	146 / 66
Nurses / Other	113 / 99
Day / Night Shift	144 / 68

II. METHODS AND MATERIALS

In this Section we describe the data used in our experiments, the features extracted, including the contextual cues, the classifiers used, and the testing setup.

A. Data collection

Data was collected from 212 participants from a pool of employees (nurses and staff) of a large urban hospital in California [23]. Approximately, two-thirds of the participants were nurses while one-third were hospital staff. Data collection protocol approval was obtained from the Institutional Review Board of the affiliated institutions and participants consented to participate in the study. During 10 weeks, participants carried out their work day as usual, but were asked to fill a brief smartphone-based daily survey that included information on different mental constructs, including stress on a 5-point scale. Study demographics are shown in Table I.

Participants were asked to wear a Fitbit charge 2 wristband at all times to collect their heart rate, sleep quality, and activity. In addition, at work, they were asked to wear a Unihertz Jelly Pro smartphone that was programmed to act as a personal audio feature recorder, termed TILES Audio Recorder (TAR) [12]. Due to privacy concerns, audio features were extracted on the device and the actual audio was discarded. An energy-based voice activity detector was implemented so features were only extracted once the user spoke. While at work, participants were also asked to wear an OMSignal smartshirt/bra that monitored their electrocardiograms, breathing patterns, and activity levels.

In addition to audio feature extraction, the smartphones were also used to send Bluetooth packets every second over 15-second windows every minute to allow for estimation of their locations within the hospital. These packets had a unique 4-bytes identifier for every participant. Owl-in-One Bluetooth data hubs were placed throughout the hospital to measure participant proximity by capturing the signal strength of Bluetooth packets. In particular, the Owl-in-Ones were installed in fourteen nursing units (spread over seven of the building floors) and two hospital labs. A total of 244 Owl-in-Ones were installed, about 1.5 m to 2.0 m above the floor depending on space availability on wall areas near power outlets. Each nursing unit was equipped with an Owl-in-One sensor in these four room types: patient room, nursing station, lounge, and medication room. All data were de-identified and uploaded daily to a secure server. More details about the dataset collection can be found in [23].

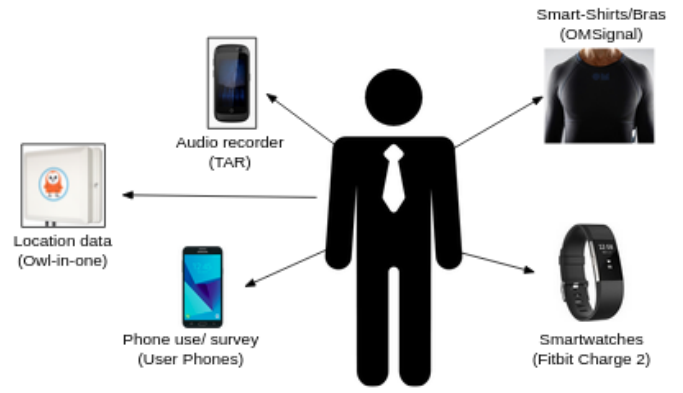


Fig. 1. Sensors used in the study

B. Feature Extraction

1) *Audio*: The TAR was programmed with two open-source toolkits: openSMILE and Tarsos-DSP. Tarsos-DSP [25] was used for voice activity detection (VAD) in order to trigger openSMILE feature extraction [26] only when the subject was speaking. A total of 28 audio features were extracted, including prosodic (e.g., pitch, intensity, voicing probability), spectral (e.g., band energy between 250 Hz to 650 Hz, centroid of frequency distribution, spectral rolloffs, intensity, loudness), and other acoustic measures (e.g., zero crossing rate, 14th order mel-frequency cepstral coefficients). Features were extracted every 25 ms with a 10 ms frame shift. In our experiments, we average all feature frames over one minute segments. When experimenting with speech-only stress detection, four statistical functionals (mean, variance, skewness, kurtosis) are computed per feature over a 1-hour window for all hours worked in a shift.

It is important to emphasize that a simple energy-based VAD was used in order to allow the TAR to run consecutively for 10 hours on a single charge. Energy-based VADs are prone to false triggers from environment sounds. While an effort was made to direct the smartphone microphone towards the mouth of the user and gains were regulated accordingly, hospitals are known to be extremely noisy environments, thus false VAD triggers are expected. To reduce the negative impact this can have on stress detection, a foreground-background speech detection module was also used to discard false triggers. The method relies on the same features described above; more details can be found in [27]. Henceforth, only speech frames that received the “foreground” label are used in our analyses.

2) *Location*: To find the location of the subject, the Owl-in-One data hub that receives the maximum received signal strength indicator (RSSI) from the subject is selected to be the nearest sensor to the health worker. This particular sensor then provides the location within the hospital for each subject for each minute of their shift.

In order to add location as context for stress detection, we aggregate audio features on an hourly basis based on one of the 244 possible locations. Four statistical measures (mean, variance, skewness, kurtosis) are then computed for all audio

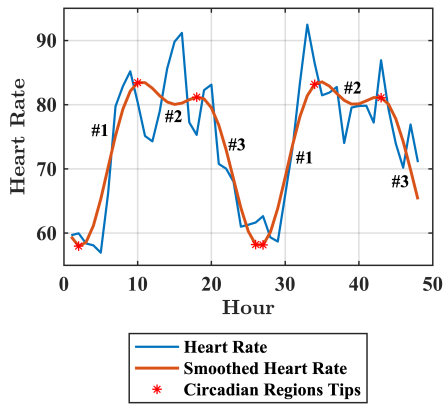


Fig. 2. Circadian rhythm during a 48 hour window of a subject

feature frames over a given hour within a specific room. This is done for all hours of a participant’s work shift. An additional feature is also computed and corresponds to the amount of time the participant spends within a workday in each location.

3) *Circadian rhythm*: It has been shown that time of stressor exposure within the day can have varying effects on individuals, in particular with disruption of their circadian rhythm [28]. Circadian rhythms vary from person to person, thus it is hypothesized that a “time of day” context may not fully capture the individualized effects that stressors may have on acoustic features. As such, we propose to aggregate speech features based on regions of the person’s circadian cycle.

Figure 2 depicts the heart rate curve of a participant over a 48-hour period, along with a moving-average smoother curve. For simplicity, we divide the daily circadian rhythm into four regions. The first corresponds to the sleep region and exhibits a low heart rate (roughly represented by the valleys in the plot). The second region corresponds to the wakefulness period in which a user is up and ready to start the day. This is characterized by an increase in heart rate. Region three corresponds to mid-day region where the heart rate is stabilized, with the occasional drop after e.g., lunch. The last region corresponds to the end of the day where the body prepares itself to enter the sleep period.

As day shift and night shift workers exhibit very different circadian rhythm curves, we decided to test our hypothesis with only day-shift workers. As such, our analysis relied on 144 subjects. In this case, the same audio feature statisticals were used, but now also exhibiting details about which circadian rhythm region the participant was in.

C. Classification

Deep neural networks have shown to provide state-of-the-art accuracy across a number of speech applications. Long short-term memory (LSTM) networks, in particular, have shown to be very useful across numerous voice applications, including speech emotion recognition [29]. LSTMs are particularly useful as they mitigate the problem of vanishing or exploding

gradients [30]. They have an internal memory cell that is controlled by forget and input gate networks [30], [31]. The forget gate controls how much of prior memory should be passed to the next time step LSTM cell, while the input gate controls how much of new input should be fed to the current time step [30]. As temporal pooling has been shown to be important for “in-the-wild” speech emotion recognition [32], [33], LSTMs should play a crucial role for the task at hand.

LSTM networks, however, can only make use of past time steps. With speech, and in particular speech stress detection, obtaining feedback from future time steps can help enhance classifier performance [31]. As such, here we rely on bi-directional LSTMs (Bi-LSTM) [24] for improved speech stress emotion detection. Figure 3 depicts the architecture of the Bi-LSTM used in our experiments.

D. Testing setup

The full dataset (144 participants $\times$ approximately 7-hour shifts $\times$ five days per week $\times$ ten weeks) was divided 80%-20% into training and test sets, respectively. To ensure the generalizability of the model, the training and test sets are comprised of different subjects. For the training data, 20% is set aside for model validation.

As mentioned previously, we are interested in exploring the effects of context on stress detection. As such, three conditions are explored here: (1) audio alone, (2) audio with location, and (3) audio with location and circadian rhythm based contextual cues. Stress ratings were reported by the participants on a 5-point scale. In order to take into account subjective variability, stress levels were transformed to z-scores per subject and separated into two classes, namely “stress” (above the mean) and “no stress” (below) days. As a result, a total of 600 “no stress” and 629 “stress” days of data are available.

For training of the Bi-LSTM, the number of epochs was set to 300, the learning rate to 0.001, and early stopping was enabled to avoid overfitting. Both the forward and backward LSTMs contained 60 neurons. The output of these two layers was then fed into a dense layer comprised of one neuron that estimates the stress level for a given day. Here, we use accuracy and F1 score as figures-of-merit.

III. EXPERIMENTAL RESULTS AND DISCUSSION

Figure 4 shows the performance of the BiLSTM for the three tested conditions: i) audio-only, ii) audio and location, and iii) audio, location and circadian rhythm. As can be seen, adding contextual information improved both accuracy and F1 scores, with the largest improvements occurring when both location and circadian rhythm information were used. These findings show the importance of including context into speech-based stress predictors for in-the-wild applications.

While speech has shown to convey important emotional information (e.g., [33]), these can be severely affected by the environment. Within a hospital setting, for example, speech collected within the cafeteria while on break versus in a patient room, or even at the emergency room can have very distinct patterns and provide different emotional cues. Our results

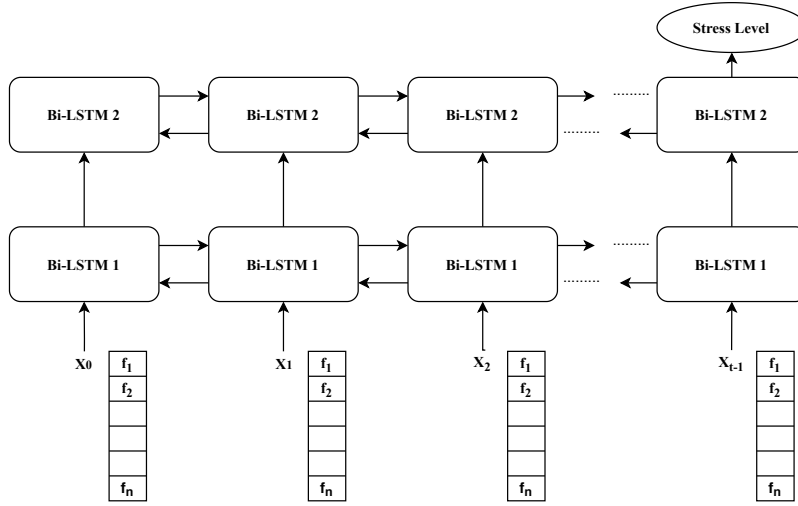


Fig. 3. Architecture of the Bi-LSTM model used for stress detection. The f_i 's correspond to feature vectors which are speech features aggregated based on time only (condition 1), time and location (condition 2), or location and time within the circadian cycle (condition 3).

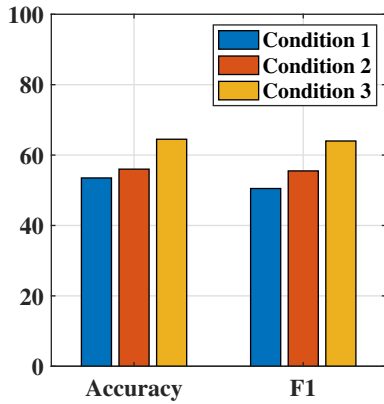


Fig. 4. Performance comparison across three conditions.

show that aggregating speech features per room resulted in an increase in accuracy (relative to speech alone) of 2.5% and 5% in F1 score. It is believed that by separating speech features per location, the network was able to better learn the impact of socialization (e.g., conversations in the break room or outdoors) [34] from stressful events in a noisy emergency room [16], thus improving overall stress detection.

Moreover, individuals have varying circadian rhythms that can be affected by sleep-wake cycles and/or stress levels. Typically, peak sympathetic system activation (fight or flight system) is reached during the early hours of the day with a dip towards the evening [35]. Detecting these changes on a day-to-day basis for each participant could provide important context for improved stress prediction. In fact, inclusion of circadian rhythm information resulted in a further increase of 8.5% in accuracy and 8.5% in F1 score relative to when audio and location were used. Here, the measured heart rates were only used to obtain a circadian curve used to aggregate

speech features based on when they occurred in the cycle. It is believed that a multimodal system that also incorporates heart rate and heart rate variability measures [36], as well as breathing patterns [37] could further improve system accuracy.

IV. CONCLUSIONS AND FUTURE WORK

This paper investigated the application of Bi-LSTM networks to detect stress in day-shift hospital worker. Models were developed based on speech features alone, as well as speech features aggregated based on location and circadian rhythm contextual cues. It was found that location improved accuracy and F1 scores by 2.5% and 5% respectively, with circadian rhythm context providing a further 8.5% and 8.5% gain. These findings show the importance of adding contextual information for stress detection, especially in hospital settings. Amidst a COVID-19 pandemic and overwhelmed hospitals, this will be crucial to assure the well-being of frontline hospital workers. For future work, analyses with both day and night shift workers should be conducted to explore the effects of circadian rhythm disruptions on stress detection. Moreover, multimodal systems based on speech and physiological parameters (e.g., heart and breathing rate) should be explored.

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